

pyUDLF: A Python Framework for Unsupervised Distance Learning Tasks

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ABSTRACT

The representation of multimedia content experienced tremendous advances in the last decades. Mainly supported by deep learning models, impressive results have been obtained. However, despite such advances in representation, the definition of similarity has been neglected. Effectively computing the similarity between representations remains a challenge. Traditional distance functions, such as the Euclidean distance, are not able to properly consider the relevant similarity information encoded in the dataset manifold. In fact, manifolds are essential to perception in many scenarios, such that exploiting the underlying structure of dataset manifolds plays a central role in multimedia content understanding and retrieval. In this paper, we present a framework for unsupervised distance learning which provides easy and uniform access to methods capable of considering the dataset manifold for redefining similarity. Such methods perform context-sensitive similarity learning based on more global measures, capable of improving the effectiveness of retrieval and machine learning tasks. The framework can use distance, similarity, or ranking information both as input and output and compute traditional retrieval effectiveness measures. Implemented as a wrapper in Python, the framework allows integration with a large number of Python libraries while keeping a back-end in C++ for efficiency. The paper also discusses diverse applications of the methods available in the pyUDLF framework, including image re-ranking, video retrieval, person re-ID, and pre-processing of distance measurements for clustering and classification.

CCS CONCEPTS

• Information systems → Retrieval models and ranking.

KEYWORDS

multimedia retrieval, unsupervised, distance learning, framework

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1 INTRODUCTION

The effectiveness of multimedia retrieval tasks relies on two pivotal factors: (i) the representation of data objects regarding their multimedia content, and (ii) the accurate measurement of their similarity considering extracted representations. Although obtaining discriminative feature representations is at the core of the task, effective distance/similarity measures are also essential for effective retrieval and machine learning applications. However, while multimedia representation approaches have experienced huge advances, mainly supported by Convolutional Neural Networks (CNNs) and Transformer-based models, distance measures have been mostly neglected.

As a result, the effective similarity definition between representations is still an open and challenging question. The main motivation consists in the way multimedia data is often modeled and compared. Images and other multimedia data can be represented in spaces of much smaller dimensions than their respective feature vectors. In this scenario, exploiting the intrinsic structure of datasets becomes a central problem in retrieval, learning, and computational vision tasks [5, 7]. The multimedia content is commonly represented as high-dimensional points in an Euclidean space and the distance between two objects is computed often considering the Euclidean distance. However, once pairwise distance measures define relationships only between pairs of images, the global structure of the dataset and the context wherein the query is computed are ignored.

In general, this is the central point of unsupervised distance learning methods, whose objective is to compute more effective distances among images, capable of taking into account the relationships among data items and the global structure of datasets. Such methods provide contextual-sensitive similarity measures, which have been successfully applied to capture the geometry of the underlying manifold on two main categories: diffusion and rank-based approaches. Diffusion processes [2, 5, 29–31] propagate similarities through graph structures generated based on feature information. They rely on the definition of a global measure, which describes the relationship between pairs of points in terms of their connectivity. The pairwise similarities are re-evaluated in the context of other images, spreading the similarity across the graph and improving the effectiveness of retrieval tasks [2, 8]. Rank-based approaches, on the other hand, provide effective yet efficient solutions [1, 3, 17, 28].

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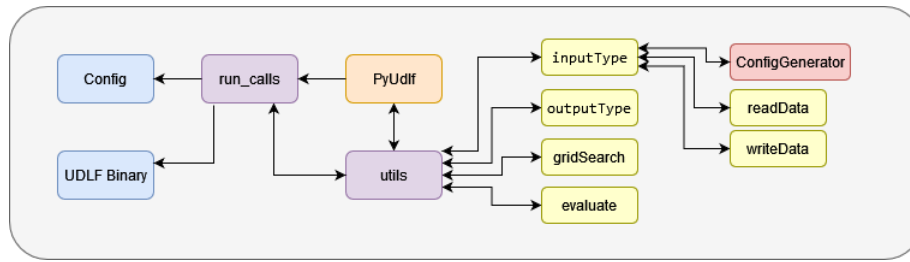


Figure 1: pyUDLF general architecture.

Ranked lists encode contextual information by establishing similarity relationships among a set of objects, which can be exploited for effective retrieval tasks [13].

In this work, we present pyUDLF: a Python framework for unsupervised distance learning methods. The framework offers a well-defined API with easy access to different unsupervised distance approaches. Currently, the framework includes 11 distinct methods based on diffusion processes, graphs, and rank-based approaches. The framework architecture is based on a back-end [23] implemented in C++ and a wrapper component in Python. In this way, the framework combines efficiency aspects with broad availability, allowing easy integration with Python machine-learning libraries. The framework is also flexible in terms of input/output data and works with distance, similarity, or rank-based information.

A discussion about the framework architecture is presented in Section 2. The methods currently available in the framework are discussed in Section 3. The diversified applications in multimedia retrieval and machine learning are presented in Section 4. Finally, Section 5 discusses the conclusions.

2 FRAMEWORK ARCHITECTURE

The architecture and main features of the framework are discussed in this section. Section 2.1 details the back-end component while Section 2.2 focuses on the Python API. The framework usage is discussed in Section 2.3. The framework is publicly available on Github¹. We also provide a zip file² containing the latest pyUDLF and UDLF source codes, datasets, and some execution examples.

2.1 UDLF Architecture

The original UDLF [23] is an open-source software implemented in C++. Currently, the framework implements several distance learning approaches and it's flexible to allow more methods to be integrated into it. There are 4 categories of classes that compose the architecture: (i) *Core* is responsible to read the execution settings from a text file and validate the parameters; (ii) *Methods* where every approach has a different class; (iii) *Evaluations* implements the necessary measures to evaluate the results; and (iv) *Utils* consists in all the auxiliary functions. There is a class named *Udl* that defines the interface with all the functions that every method is expected to implement. A new method can be implemented by inheriting from *Udl* class.

2.2 pyUDLF Architecture

The pyUDLF framework was designed to provide easy accessibility to unsupervised distance learning methods while keeping efficiency

and flexibility. The method executions are performed in the back-end and the Python component aims to provide easy API access. In this way, the input/output interface can be done through files or using Python memory structures.

Figure 1 illustrates the general pyUDLF architecture. The pyUDLF framework is composed of two modules: *run_calls* and *utils*. The *run_calls* is responsible for running the UDLF binary and processing the input and output files. It allows the execution of the UDLF with different settings. The *utils* module is divided into four parts: *InputType*, *OutputType*, *GridSearch*, and *Validation*. *InputType* handles the reading, writing, and configuration of the UDLF input files, allowing the manipulation of the data needed for execution. *OutputType* deals with the output files generated by the UDLF and visualization purposes. *GridSearch* enables grid searching of parameters in order to find the best configuration. *Validation* performs the calculation of effectiveness measures such as Precision, Recall, and MAP (Mean Average Precision).

2.3 Framework Usage

An example of framework usage in Python is depicted in Listing 1. The example considers files for input/output.

```
1 from pyUDLF import run_calls as udlf
2 from pyUDLF.utils import inputType
3 # Paths for the binary and the configuration file.
4 udlf.setBinaryPath('../UDLF/bin/udlf')
5 udlf.setConfigPath('../UDLF/bin/config.ini')
6 # Path to the ranked list file
7 RksPath = "../Dataset/ranked_list/rks.txt"
8 # Path to the lists file
9 pListFile = "../Dataset/lists.txt"
10 # Path to the classes file
11 pClassesFile = "../Dataset/classes.txt"
12 # Dataset Size
13 pN = 1360
14 # Creating an instance of InputType
15 input_data = inputType.InputType()
16 # Setting parameters for input_data
17 input_data.set_param("SIZE_DATASET", pN) # Dataset size
18 input_data.set_ranked_lists_size(pN) # Ranked list size
19 input_data.set_method_name("CPRR") # Method name
20 input_data.set_input_files(RksPath) # Input files
21 input_data.set_lists_file(pListFile) # List file
22 input_data.set_classes_file(pClassesFile) # Classes file
23 # Running UDLF and getting results
24 output = udlf.run(input_data, get_output=True)
25 # Printing the log from the output
26 output.print_log()
27 # Getting the log values from the output
28 return_values = output.get_log()
```

Listing 1: pyUDLF execution example of CPRR method.

¹Repository: <https://github.com/UDLF/pyUDLF>

²Zip file: <http://pyudlf.lucasvalem.com>

3 UNSUPERVISED DISTANCE LEARNING METHODS

The pyUDLF framework currently has 11 different unsupervised distance learning methods available. A common characteristic among all the methods consists in the use of rank-based approaches, which represent a crucial strategy for efficiency aspects, once the most relevant similarity information is encoded at the top positions of ranked lists. The methods available are briefly presented in this section, following an inverse chronological order of proposition:

- **RFE [27]**: the Rank Flow Embedding algorithm computes contextual-sensitive embeddings through a rank-based processing flow. The method combines hypergraphs, Cartesian products, and connected components in order to maximize the similarity information encoded in the embeddings.
- **BFS-Tree [17]**: a Breadth-First Tree is employed to represent the similarity information derived from ranking references. The tree structure is then utilized to uncover underlying similarity relationships, giving rise to a more effective distance measure.
- **RDPAC [18]**: the Rank Diffusion with Assured Convergence method provides an efficient approximation of a diffusion process using rank-based information. The approach combines the advantages of traditional diffusion process methods while keeping efficiency by using only the top positions of the ranked list.
- **LHRR [16]**: the Log-based Hypergraph of Ranking References method uses a hypergraph for representing similarity information encoded in ranked lists. Subsequently, the similarity between hyperedges is used to derive a more effective similarity measure.
- **Correlation Graph [14]**: the method exploits a correlation graph based on high-confidence edges between very similar nodes. Subsequently, the similarity is spread through the graph using its strongly connected components.
- **RL-Sim [13, 22]**: the Ranked List-Similarities method is based on the conjecture that the ranked lists encode contextual similarity information which can be used to compute a more effective distance between objects. The algorithm employs an iterative re-ranking approach based on rank correlation measures.
- **CPRR [25]**: the Cartesian Product of Ranking References (CPRR) method employs Cartesian product operations computed over the k NN and reverse k NN sets of dataset images. In this way, such operations extract relevant similarity information from sets defined by ranked lists.
- **Ranked List Graph Distance [11]**: in this approach, each ranked list is represented as a weighted sub-graph, with the weights determined by the top- k positions. These sub-graphs are then merged into a single graph, where the edge weights are utilized to increase the similarity scores between images.
- **RL-Recom [28]**: the method capitalizes on the notion of recommendations within ranked lists. The underlying motivation stems from the conjecture that the similarity information present in the ranked lists can be leveraged to suggest images to one another.
- **Manifold Reciprocal kNN Graph [15]**: the method constructs a graph by taking into account the reciprocal references found within the top- k positions of the ranked lists. By leveraging the inherent structure of the graph, the method utilizes scores to analyze the manifold of the dataset and generate new ranked lists.
- **ContextRR [12]**: an image structure derived from ranked lists and distance measures is exploited to represent contextual similarity encoded in the dataset. Leveraging the information extracted from these context images, a refined distance measure is computed.

4 MULTIMEDIA RETRIEVAL AND MACHINE LEARNING APPLICATIONS

The methods available in the pyUDLF framework have been exploited in diversified multimedia and machine-learning scenarios. This section discusses applications in different fields and tasks.

4.1 Image and Video Retrieval

Image and video retrieval tasks greatly benefit from the information contained in the dataset manifold, especially when combined with representative features. Most of the unsupervised distance learning methods [14, 16–18, 22, 27] were evaluated on image retrieval applications, considering different features. Additionally, various methods can also be used in rank aggregation, allowing for combination of different descriptors.

The *BFS-Tree* is used to improve image retrieval in [17]. The similarity extracted from the tree-based algorithm was able to improve the separability among classes. Figure 2 presents the results obtained in the MPEG-7 dataset. Three reference images were selected and used to retrieve multiple images from the data, based on dissimilarity. The red dots represent images relevant to the reference images. After the *BFS-Tree* execution, the relevant and non-relevant images are clearly separated in the space.

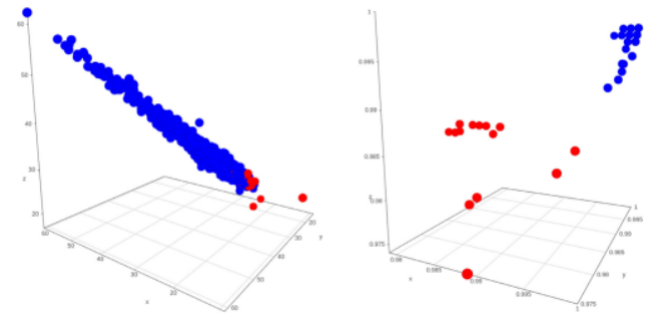


Figure 2: BFS-Tree impact on class-separation on selected images from MPEG-7 dataset before and after the application of the algorithm. Left and right plots, respectively [17].

Finally, a multimodal video retrieval algorithm is proposed in [4] combining rank aggregation, based on *LHRR* and *BFS-Tree*, and Graph Convolutional Networks (GCNs). The method explores the similarity ranking of each modality separately, uniting them with rank aggregation for generating inputs for training a GCN. The output embedding is used for the final retrieval task. Additionally, two different approaches are proposed, processing the features combined in a single GCN or separately, with the addition of a final GCN based on the combined input of the earlier ones.

4.2 Person Re-ID

Person Re-ID is a complex task that aims to classify and retrieve multiple views of a single individual. In [24], an unsupervised approach is proposed by combining multiple different descriptors using hypergraph techniques for rank selection, fusion, and aggregation.

Moreover, *RDPAC* is used in [18] for the Person Re-ID task, obtaining large gains over the original features ranking. Figure 3 presents an example of this application. The obtained rank improves the results, only missing one of the retrieved images.



Figure 3: Person Re-ID results, before and after use of unsupervised distance learning approaches [18].

4.3 Weakly Supervised Classification

The extraction of new information and similarities from manifold learning techniques can also improve weak-supervised learning tasks, where the availability of labeled data is limited. In these scenarios, improving the relationship between the data with unsupervised distance learning techniques helps to extract better results from small labeled sets.

A hypergraph-based approach for labeled set expansion is proposed in [20]. The approach explores *LHRR* formulation to select reliable unlabeled candidates, expanding the available training set and achieving better results in the weakly-supervised task. In [26], manifold learning techniques are applied to create kNN-graphs used in a novel *manifold-GCN* architecture. The application of re-ranking techniques before weak-supervised training was able to improve results in most evaluated situations. Additionally, the framework allows the use of different re-ranking and GCN architectures, being able to achieve effective results in different scenarios.

4.4 Clustering

Clustering is a traditional unsupervised task that aims to separate data into different groups, based on the relationship of their representations. In this scenario, distance learning techniques can be used to greatly improve clustering results in multiple ways. A framework based on applying manifold learning followed by clustering techniques is proposed in [21]. Figure 4 presents obtained results on the classic Spirals dataset from three different clustering techniques before and after the application of manifold learning. In all images, the obtained separation is improved by the applied techniques, being able to perfectly separate the data in two scenarios.

Additionally, manifold learning can also be applied to derive new clustering techniques. In [10], a new method is proposed based on the discovery of connected components in a reciprocal threshold-based kNN graph. Moreover, in [9], *LHRR* is applied to recover better graphs and to select reliable soft-labels, applied to train a GCN for clustering the data, in a self-supervised manner.

4.5 Other Machine Learning Applications

This section does not intend to provide an exhaustive list of applications, but rather to present an overview of already exploited applications and potential use cases. In another distinct scenario, unsupervised distance learning methods have already been exploited for regression tasks. In [6], a regression by re-ranking approach

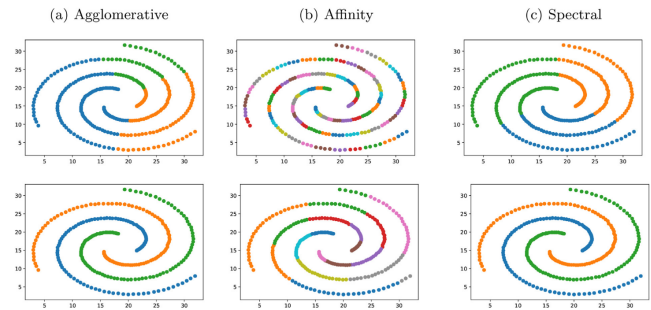


Figure 4: Impact on clustering results: before and after distance learning, on first and second lines, respectively [21].

is proposed and validated in general regression datasets and brain age prediction applications. The capacity of encoding contextual information is also exploited for feature augmentation in [19], in order to achieve better image classification results. The methods are also used in anomaly detection [10] applications, using graph and rank-based approaches.

5 CONCLUSIONS

Similarity measurement is a crucial task, ubiquitous in many multimedia retrieval and machine learning activities. Traditionally computed by pairwise distance measures, more global similarity relationships are ignored. Contextual-sensitive similarity measures represent a fundamental solution by taking into account the dataset manifold in order to improve effectiveness. In this scenario, software tools that can make such methods easily accessible are of paramount importance. We contribute to bridging this gap by proposing a Python framework with various recent unsupervised distance learning methods available.

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REFERENCES

- [1] S. Bai and X. Bai. 2016. Sparse Contextual Activation for Efficient Visual Re-Ranking. *IEEE Transactions on Image Processing* 25, 3 (March 2016), 1056–1069.
- [2] Song Bai, Xiang Bai, Qi Tian, and Longin Jan Latecki. 2019. Regularized Diffusion Process on Bidirectional Context for Object Retrieval. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 41, 5 (2019), 1213–1226.
- [3] Xiang Bai, Song Bai, and Xinggang Wang. 2015. Beyond diffusion process: Neighbor set similarity for fast re-ranking. *Information Sciences* 325 (2015), 342–354.
- [4] Lucas Barbosa de Almeida, Lucas Pascotti Valem, and Daniel Carlos Guimarães Pedronette. 2022. Graph Convolutional Networks and Manifold Ranking for Multimodal Video Retrieval. In *2022 IEEE International Conference on Image Processing, ICIP*. 2811–2815.
- [5] M. Donoser and H. Bischof. 2013. Diffusion Processes for Retrieval Revisited. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 1320–1327.
- [6] Filipe Marcel Fernandes Gonçalves, Daniel Carlos Guimarães Pedronette, and Ricardo da Silva Torres. 2023. Regression by Re-Ranking. *Pattern Recognit.* 140 (2023), 109577.
- [7] Jiayan Jiang, Bo Wang, and Zhuowen Tu. 2011. Unsupervised metric learning by Self-Smoothing Operator. In *International Conference on Computer Vision (ICCV)*. 794–801.
- [8] Chundi Liu, Guangwei Yu, Maksims Volkovs, Cheng Chang, Himanshu Rai, Junwei Ma, and Satya Krishna Gorti. 2019. Guided Similarity Separation for Image Retrieval. In *Advances in Neural Information Processing Systems*, Vol. 32.
- [9] Leonardo Tadeu Lopes and Daniel Carlos Guimarães Pedronette. 2023. Self-Supervised Clustering Based on Manifold Learning and Graph Convolutional Networks. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 5634–5643.
- [10] Leonardo Tadeu Lopes, Lucas Pascotti Valem, Daniel Carlos Guimarães Pedronette, Ivan Rizzo Guilherme, Joao Paulo Papa, Marcos Cleison Silva Santana, Danilo Colombo, GM Farinella, P Radeva, and J Braz. 2020. Manifold Learning-based Clustering Approach Applied to Anomaly Detection in Surveillance Videos. In *Proceedings Of The 15th International Joint Conference On Computer Vision, Imaging And Computer Graphics Theory And Applications, Vol 5: Visapp*. Scitepress, 404–412.
- [11] Daniel Carlos Guimarães Pedronette, Jurandy Almeida, and Ricardo da S. Torres. 2016. A graph-based ranked-list model for unsupervised distance learning on shape retrieval. *Pattern Recognition Letters* 83, Part 3 (2016), 357–367.
- [12] Daniel Carlos Guimarães Pedronette and Ricardo da S. Torres. 2012. Exploiting Contextual Information for Image Re-Ranking and Rank Aggregation. *International Journal of Multimedia Information Retrieval* 1, 2 (2012), 115–128.
- [13] Daniel Carlos Guimarães Pedronette and Ricardo da S. Torres. 2013. Image Re-Ranking and Rank Aggregation based on Similarity of Ranked Lists. *Pattern Recognition* 46, 8 (2013), 2350–2360.
- [14] Daniel Carlos Guimarães Pedronette and Ricardo da S. Torres. 2016. A correlation graph approach for unsupervised manifold learning in image retrieval tasks. *Neurocomputing* 208 (2016), 66–79.
- [15] Daniel Carlos Guimarães Pedronette, Otávio A.B. Penatti, and Ricardo da S. Torres. 2014. Unsupervised manifold learning using Reciprocal kNN Graphs in image re-ranking and rank aggregation tasks. *Image and Vision Computing* 32, 2 (2014), 120–130.
- [16] Daniel Carlos Guimarães Pedronette, Lucas Pascotti Valem, Jurandy Almeida, and Ricardo da Silva Torres. 2019. Multimedia Retrieval Through Unsupervised Hypergraph-Based Manifold Ranking. *IEEE Trans. Image Process.* 28, 12 (2019), 5824–5838.
- [17] Daniel Carlos Guimarães Pedronette, Lucas Pascotti Valem, and Ricardo da Silva Torres. 2021. A BFS-Tree of ranking references for unsupervised manifold learning. *Pattern Recognit.* 111 (2021), 107666.
- [18] Daniel Carlos Guimarães Pedronette, Lucas Pascotti Valem, and Longin Jan Latecki. 2021. Efficient Rank-Based Diffusion Process with Assured Convergence. *J. Imaging* 7, 3 (2021), 49.
- [19] Vanessa Helena Pereira-Ferrero, Lucas Pascotti Valem, and Daniel Carlos Guimarães Pedronette. 2023. Feature augmentation based on manifold ranking and LSTM for image classification. *Expert Syst. Appl.* 213, Part (2023), 118995.
- [20] João Gabriel Camacho Presotto, Samuel Felipe dos Santos, Lucas Pascotti Valem, Fábio Augusto Faria, João Paulo Papa, Jurandy Almeida, and Daniel Carlos Guimarães Pedronette. 2022. Weakly supervised learning based on hypergraph manifold ranking. *J. Vis. Commun. Image Represent.* 89 (2022), 103666.
- [21] Bionda Rozin, Vanessa Helena Pereira-Ferrero, Leonardo Tadeu Lopes, and Daniel Carlos Guimarães Pedronette. 2021. A rank-based framework through manifold learning for improved clustering tasks. *Inf. Sci.* 580 (2021), 202–220.
- [22] Lucas Pascotti Valem, Carlos Renan De Oliveira, Daniel Carlos Guimarães Pedronette, and Jurandy Almeida. 2018. Unsupervised Similarity Learning through Rank Correlation and kNN Sets. *ACM Trans. Multim. Comput. Commun. Appl.* 14, 4 (2018), 80:1–80:23.
- [23] Lucas Pascotti Valem and Daniel Carlos Guimarães Pedronette. 2017. An Unsupervised Distance Learning Framework for Multimedia Retrieval. In *ACM on International Conference on Multimedia Retrieval, ICMR*. 107–111.
- [24] Lucas Pascotti Valem and Daniel Carlos Guimarães Pedronette. 2022. Person Re-ID through unsupervised hypergraph rank selection and fusion. *Image Vis. Comput.* 123 (2022), 104473.
- [25] Lucas Pascotti Valem, Daniel Carlos Guimarães Pedronette, and Jurandy Almeida. 2018. Unsupervised similarity learning through Cartesian product of ranking references. *Pattern Recognition Letters* 114 (2018), 41–52. Data Representation and Representation Learning for Video Analysis.
- [26] Lucas Pascotti Valem, Daniel Carlos Guimarães Pedronette, and Longin Jan Latecki. 2023. Graph Convolutional Networks based on manifold learning for semi-supervised image classification. *Comput. Vis. Image Underst.* 227 (2023), 103618.
- [27] Lucas Pascotti Valem, Daniel Carlos Guimarães Pedronette, and Longin Jan Latecki. 2023. Rank Flow Embedding for Unsupervised and Semi-Supervised Manifold Learning. *IEEE Trans. Image Process.* 32 (2023), 2811–2826.
- [28] Lucas Pascotti Valem, Daniel Carlos Guimarães Pedronette, Ricardo da S. Torres, Edson Borin, and Jurandy Almeida. 2015. Effective, Efficient, and Scalable Unsupervised Distance Learning in Image Retrieval Tasks. *ACM International Conference on Multimedia Retrieval (ICMR)* (2015).
- [29] Xingwei Yang, Xiang Bai, Longin Jan Latecki, and Zhuowen Tu. 2008. Improving Shape Retrieval by Learning Graph Transduction. In *European Conference on Computer Vision (ECCV)*, Vol. 4. 788–801.
- [30] Xingwei Yang, Suzan Koknar-Tezel, and Longin Jan Latecki. 2009. Locally constrained diffusion process on locally densified distance spaces with applications to shape retrieval. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 357–364.
- [31] Xingwei Yang, L. Prasad, and L.J. Latecki. 2013. Affinity Learning with Diffusion on Tensor Product Graph. *IEEE Trans. on Pattern Analysis and Machine Intelligence* 35, 1 (2013), 28–38.